

JEREMY THUMMEL

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EDUCATION

Ontario Tech University | Oshawa, ON

Bachelor's in Computer Science (Honours) | September 2022 - May 2027

GPA: 4.06 / 4.30 | President's List Member, 2022 - 2026

Relevant Courses: Machine Learning Theory and Applications, Machine Learning II, Natural Language Processing, Data Structures, Analysis and Design of Algorithms, Database Systems & Concepts, Computer Architecture, Linear Algebra, Discrete Mathematics

WORK EXPERIENCE

AI Systems Developer | May 2026 - September 2026

AFM Industries | Whitby, ON

- Developed and deployed a conversational LangGraph application in Microsoft Teams to help employees resolve company operations involving industrial belting, achieving an 86% self-service resolution rate.
- Built a retrieval-augmented generation pipeline that ingested SharePoint documents and PostgreSQL data into Azure AI Search, using OCR, text-coordinate extraction, and image transcription to make both textual and visual content searchable.
- Implemented deterministic, state-based LangGraph workflows with fillable forms, context compaction, dynamic system prompts, supervisor agents, and judge nodes to support calculations, decision-tree logic, multimodal requests, and quote generation.
- Applied Microsoft security trimming and access-control lists to preserve source-level permissions and support secure company-wide adoption, while developing Teams and email workflows that routed state-triggered escalations and approval requests to management.
- Produced biweekly feedback reports that surfaced negatively rated interactions and summarized recurring conversation patterns.
 - Patterns were determined by grouping text according to vector similarity, and extracting categories using LLM analysis.

Computer Science Intern | January 2025 - December 2025

Teknion Furniture Systems | North York, ON

- Built a Google Apps Script and JavaScript label-printing application with a configurable HTML interface, supporting multiple manufacturing plants, printer configurations, and label formats to streamline internal printing workflows.
- Designed a centralized SQL schema to organize employee access requirements and simplify provisioning for software permissions, applications, and internal company resources.
- Integrated Zendesk-triggered Okta Workflows with internal applications using REST APIs, custom connectors, and JSON transformations, automating permission updates and reducing average provisioning time by 38%.
- Developed automated PowerShell tests to validate authentication, network access controls, and connectivity across Citrix-hosted environments, local endpoints, and internal networks.
- Created PowerShell reconciliation scripts to group and display employee data, compare it against company records, and identify access, provisioning, and data discrepancies.

Technical Director | October 2024 - April 2025

Google Developer Groups on Campus | Oshawa, ON

- Developed, presented, and published workshops reviewing the architecture and user capabilities of Google's NotebookLM platform, and the integration of Cohere language models through REST API calls in Python.

PROJECTS

M.A.G.E. - Masked and Generative Image Editor | January 2026 - April 2026

- Built a Python and Gradio image editor that applied text-guided changes to selected image regions, giving the user the option of using manual brush masks or YOLO-generated segmentation masks.
- Integrated T5 prompt enhancement with Stable Diffusion inpainting and evaluated generated edits with CLIP and SSIM to compare semantic alignment and structural similarity.

Product In-Stock Tracker | August 2025 - December 2025

- Built a Python monitoring service with Selenium WebDriver to scan configurable product pages, parse links and HTML content, filter matching inventory, and detect newly listed items.
- Deployed the Python monitoring service on a Raspberry Pi for continuous scheduled checks and configured SMTP email alerts for newly available products.

Emojiable - Emoji Generation Model | September 2024 - December 2024

- Collaborated in a four-person team to train a PyTorch DCGAN with custom generator and discriminator networks, transposed convolutions, batch normalization, and LeakyReLU activations on a multi-platform emoji dataset.

ChatNow - Chat Room Platform | April 2024 - May 2024

- Collaborated in a four-person team to build a Maven/Jakarta EE chat platform using Java WebSockets, JSON, and JavaScript, supporting real-time messaging, user sessions, dynamic rooms, chat history, and customizable themes.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, PowerShell / Bash Scripting, SQL, C++, HTML / CSS, ASM

Backend and Automation: REST APIs, WebSockets, PowerShell, JSON, Selenium, SMTP, Google Apps Script, LangGraph, PyTorch, Gradio, OpenAI APIs, Cohere APIs, Pydantic, TanStack, Microsoft Azure, Microsoft Foundry, Azure AI Search, Okta Workflows

Databases and Analysis: SQL, PostgreSQL, pgvector, Pandas, NumPy, Jupyter Notebook, Power BI, Supabase, Matplotlib